

Ray (Ruitao) Chen

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WORK EXPERIENCES

NYU Future Reality Lab

New York City, NY

Graduate Research Assistant (On-Site, with Prof. Kenneth Perlin)

January 2025 – Now

- Developed a web-based **XR user interface and a 6-channel WebSocket communication layer** connecting to the ROS2 backend for multi-sensor control and navigation planning of a drone swarm (3-5), in collaboration with the Agile Robotics and Perception Lab (work **accepted for publication** at IEEE ICRA 2026).
- Prototyped and prompt engineered (Gemini Studio API) a **real-time AI chatbot** integrating into online meeting platform *Bici*, enabling users to use natural language to manipulate 3D scene animations from generated executable JSON control code.
- Designed interactive Web-based **VR lecture experiences for NYU Virtual Reality course** (CSCI-GA 3033), using customized 3D graphics framework to enable real-time, immersive interaction in both teaching & learning.

Spatial Front, Inc.

Washington, D.C.

Data Engineer Intern in Quality Assurance (Remote)

July 2023 – August 2023

- Performed **front end API tests in the GIS** (Geographic Information System) developing environment using Postman and Swagger UI. Compiled **20+ test cases** and uploaded on Jira Platform to track display issues of the web-interface.
- Determined the performance of newly developed features in the company's database system utilizing K-means clustering via Azure Studio, with **10+ noisy data sets** being discovered.
- Participated in bi-weekly sprint cycles, contributing to feature design discussions, task scoping, peer code reviews, and release retrospectives within an Agile development environment.

7thOnline, Inc.

New York City, NY

Software Engineer Intern (On-Site)

June 2022 – August 2022

- Assisted SaaS (Software as a Service) customer support team in adjusting functionalities of software programs “Wholesale, DTC, and 7thOnline” to satisfy the needs of client companies. Increased the workflow efficiency of clients by **15%**.
- Optimized the Case-Based Reasoning algorithm** with Dr. Saman Hong to improve the performance of existing SaaS software, increased the prediction precision of wholesale planning by **10%**.
- Compiled **Java test cases** to examine data visualization of the company’s beta application “7thLite” on both PC and mobile platforms, wrote **20+ Bug Reports**, and uploaded them to the internal Red Hat bug reporting system to increase the stability.

EDUCATION

New York University

New York City, NY

Master of Science (M.S.) in Computer Science **GPA: 3.53 / 4.00**

January 2024 – May 2026

The Ohio State University (Graduated with Magna Cum Laude latin honor)

Columbus, OH

Bachelor of Science (B.S.) in Mathematics & Minor in Computer Science **GPA: 3.67 / 4.00**

January 2020 – December 2023

Related Courses: Math: Calculus, Linear Algebra, Discrete Math, Probability, Scientific Computing, Cognitive Modeling

Computer Science: Data Structures, Algorithm, Operating System, Database, Machine Learning, Virtual Reality, Computer Vision

PROJECTS

Multi-Agent Spatial Reasoning XR System

September 2025 – Now

- Architected a **multi-agent (5) collaborative backend** (Parser, Asset, Scene, Verification, Memory) using LangGraph and FastAPI, translating natural language commands into precise 3D spatial manipulations with iterative refinement loops.
- Built a **full-stack WebXR interface** (Three.js, Meta Immersive Web SDK) with real-time WebSocket synchronization, enabling users to construct and manipulate XR scenes from scratch via natural language (work targeting IEEE VR 2027).

Augmented Reality Drone Control Simulator

January 2025 – May 2025

- Designed a **full-stack AR drone teleoperation system**, combining spatial user interfaces and live hardware control.
- Built custom 3D interaction widgets using Three.js, with a Node.js backend enabling real-time synchronization across server and headset, multi-client state sharing, and low-latency command streaming to a physical drone.

Webcam-Based Gesture-Controlled Drone Interface

January 2024 – May 2024

- Designed a **real-time computer vision pipeline** integrating hand landmark extraction (MediaPipe, OpenCV) with TensorFlow Lite classification (KeyPoint + LSTM-based Point History models) to enable gesture-driven navigation of a DJI Tello drone.

SKILLS

Programming Languages: Java, Python, C++, SQL, JavaScript, HTML.

Software: VS Code, Colab, Azure Studio, MySQL, Docker, Postman, Linux, ROS2.

Skills: End-to-End testing, Full-Stack Development, Distributed System, WebXR Prototyping, Agent Architecture Design.

Interest: Classical Vocal (4-year vocal training experience, won 2nd prize in Asian Teenage Singing Festival).